Iloilo City, Philippines
Github
Portfolio

Axel John E. Nuqui

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Aspiring full-stack software engineer and CS student passionate about building AI-powered applications, scalable systems, and real-world mobile and web experiences.

Technologies and Languages

- Languages: TypeScript, Python, JavaScript, Dart, Java, C++
- Technologies: React, NextJS, React Native, Flutter, TailwindCSS, REST APIs, SQLite, MariaDB, Convex
- AI/ML: TensorFlow, PyTorch, YOLOv8, MobieNetV3, Ollama, AI Engineering
- Other: Data Structures & Algorithms, Object-Oriented Programming, Database Design, Git/GitHub, UI/UX Design, System Architecture, API Development

Projects

VEC-PRO — AI-Powered Mosquito Surveillance & Forecasting System
Thesis Project · React Native · Next.js · Convex · Clerk · MobileNetV3 · YOLOv8 · Agent-Based Modeling
Developed an end-to-end mosquito surveillance system with community reporting, AI-assisted image classification, human verification workflows, and spatiotemporal spread forecasting.

- **AEDEX (Mobile):** Built a React Native app with Clerk auth and Convex backend for community users to submit potential breeding-site reports with images.
- **AEDIFY (Web):** Developed a Next.js platform with custom Convex admin auth for barangay officials to verify reports, assign tasks to tanods, and manage risk maps.
- **AI Pipeline:** Designed and trained a MobileNetV3 + YOLOv8 multi-staged pipeline to classify images and detect breeding indicators from community submissions.
- **Forecasting:** Implemented Agent-Based Modeling incorporating weather and wind data to simulate potential mosquito spread from unresolved breeding sites.

SYNT-ASK – Multiplayer NLP-Based Impostor Detection Game
AI / Machine Learning · Python · React · Typescript · NLP · DistilBERT · Socket.IO

- Developed an AI-powered multiplayer social deduction game that uses NLP and semantic similarity analysis to identify suspicious player responses.
- Implemented a Siamese DistilBERT pipeline that generates semantic embeddings and uses cosine similarity with Leave-One-Out centroid analysis to identify semantic outliers.
- Built the NLP model training pipeline using PyTorch and Transformers, including pair-based similarity learning, Cosine Embedding Loss, validation metrics, and confusion-matrix evaluation.
- Integrated the NLP system with a real-time multiplayer architecture using Flask, Flask-SocketIO, React, TypeScript, and Socket.IO for synchronized game sessions, player submissions, and chat. - Achieved 92% accuracy, precision, recall, and F1-score during model validation.

MEMOWEE — CNN-Driven Face & Speech Recognition for Memory Support
Python · TensorFlow · CNN · Frontend (HTML, CSS, JS) · Server (Flask)

- Developed an AI-powered memory support system for dementia patients combining face recognition and speech-to-text capabilities.
- Built a Convolutional Neural Network (CNN) for real-time face identification, extracting facial embeddings to recognize and differentiate individuals under varying lighting, poses, and expressions.
- Designed the system to support memory recall by pairing visitor identification with conversation summaries, enabling patients to remember recent interactions.

Work Experience

Call Center Agent **iXL Solutions Philippines** **May 2023 – July 2023**
Inbound and 24HF Skills Iloilo City, Philippines

- Handled high-volume inbound customer interactions, troubleshooting issues and providing solutions based on established processes.
- Navigated multiple software systems to access information, document interactions, and resolve customer requests accurately.
- Collaborated with team members to escalate complex issues while consistently meeting quality and performance standards.

Bartender **Bugtong Bato Beach Resort** **March 2021 – April 2023**
Beach Resort Team San Joaquin, Iloilo, Philippines

- Delivered fast, accurate service in a high-volume beach resort environment, handling up to 50+ customer orders per shift during peak seasons.
- Communicated effectively with guests and coordinated with kitchen and service teams to ensure timely order fulfillment.
- Maintained bar inventory and ensured stock availability, reducing shortages during high-demand periods.
- Resolved customer concerns promptly, contributing to positive guest feedback and repeat visits.

Education and Certifications

Bachelor of Science in Computer Science *Expected Graduation: July 2027*
West Visayas State University – Main Campus | Iloilo City, Philippines
Relevant Coursework: Data Structures & Algorithms · Database Systems · Software Engineering · Artificial Intelligence · Machine Learning · Web Development
Thesis: VEC-PRO — Multi-Staged Deep Learning Pipeline with Agent-Based Spatiotemporal Mosquito Spread Forecasting

Interests

- Building side projects and experimenting with new technologies
- Applying AI models to real-world systems and building AI-powered applications
- Backend architecture, distributed systems, and scaling complex applications